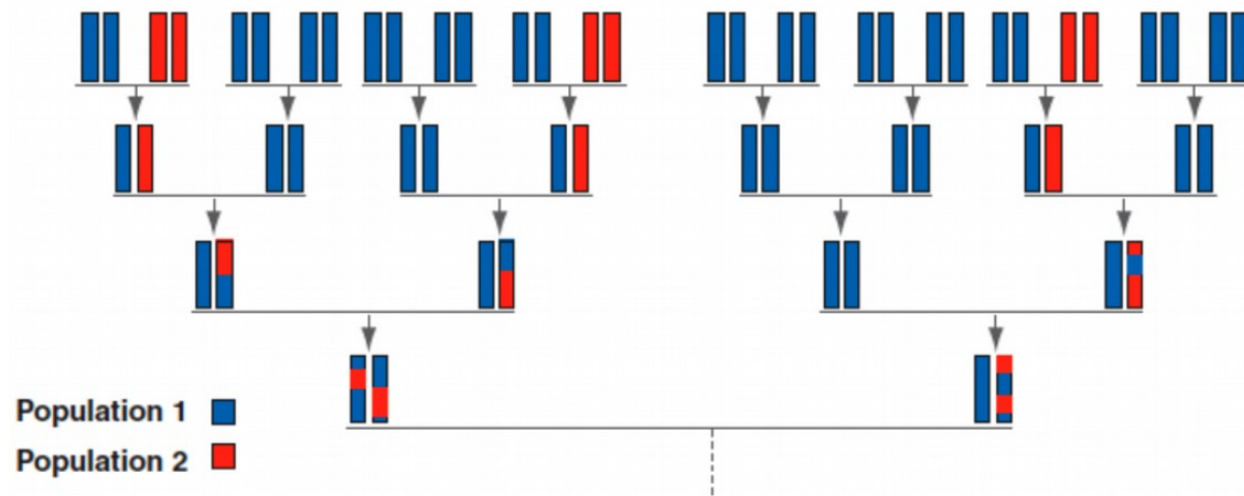


# Admixture: underlying assumptions



- genetic clustering algorithm
- at least two unadmixed populations
- at least one admixed population
- Maximum Likelihood: ADMIXTURE
- Bayesian: STRUCTURE
- delivers admixture proportions  $Q$
- delivers allele frequencies  $F$  for all loci for all  $K$  populations

# Q and F outputs of admixture analysis

**G (genotypes):**

|       |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|
| Ind 1 | A | T | G | T | T | A | A | T |
|       | T | T | G | C | T | G | T | T |
| Ind 2 | T | T | C | T | T | G | A | G |
|       | T | G | C | T | A | G | T | T |
| ...   | T | G | G | T | T | G | A | G |
|       | T | T | C | C | T | G | T | T |
|       | A | T | G | T | T | A | A | T |
|       | T | T | G | T | T | G | T | T |
|       | T | T | G | T | A | G | A | G |
|       | T | G | G | T | T | G | T | G |

SNPs

**Q (admixture proportions):**

|       |       |       |
|-------|-------|-------|
| Ind 1 | 4/16  | 12/16 |
| Ind 2 | 6/16  | 10/16 |
| ...   | 3/16  | 13/16 |
|       | 11/16 | 5/16  |
|       | 3/16  | 13/16 |

Pop1 Pop2

**F (allele frequencies):**

|      |     |     |     |     |     |     |     |     |
|------|-----|-----|-----|-----|-----|-----|-----|-----|
| Pop1 | 2/5 | 3/4 | 4/4 | 3/3 | 3/3 | 0/3 | 2/3 | 0/2 |
| Pop2 | 0/5 | 4/6 | 3/6 | 5/7 | 5/7 | 2/7 | 3/7 | 6/8 |

SNPs

# NGSadmix vs. ADMIXTURE

## JOURNAL ARTICLE

### Estimating Individual Admixture Proportions from Next Generation Sequencing Data

Line Skotte, Thorfinn Sand Korneliussen, Anders Albrechtsen [Author Notes](#)

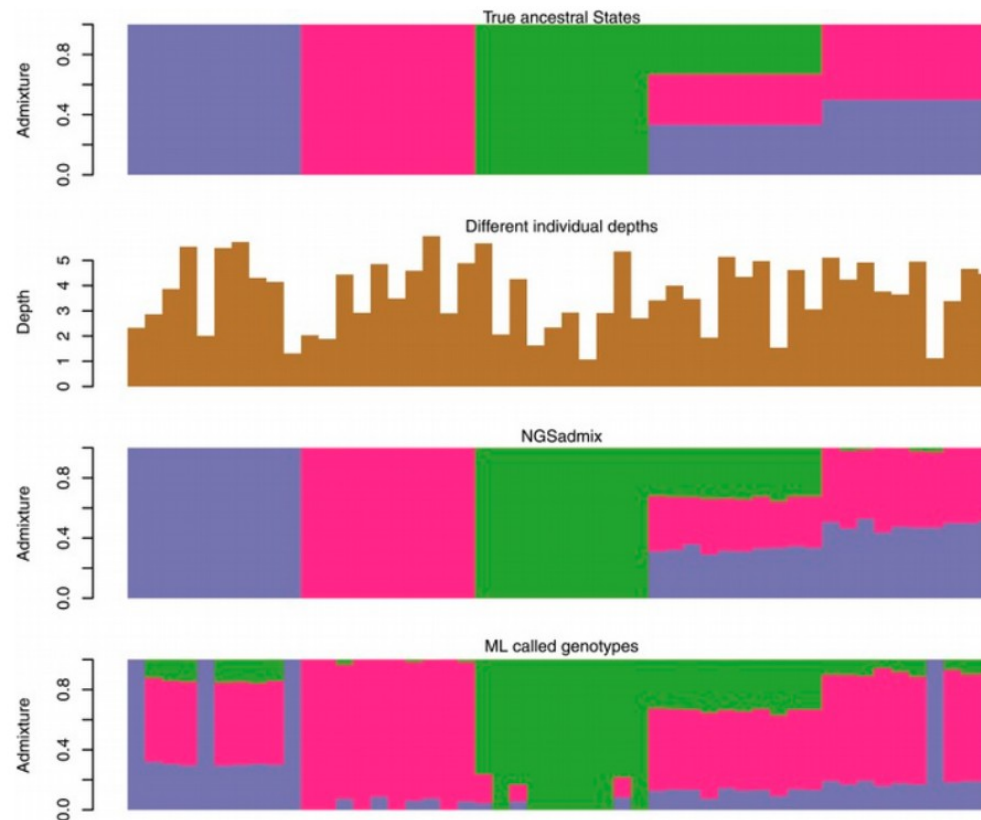
*Genetics*, Volume 195, Issue 3, 1 November 2013, Pages 693–702, <https://doi.org/10.1534/genetics.113.154138>

**Published:** 01 November 2013 [Article history](#)

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#### Abstract

Inference of population structure and individual ancestry is important both for population genetics and for association studies. With next generation sequencing technologies it is possible to obtain genetic data for all accessible genetic variations in the genome. Existing methods for admixture analysis rely on known genotypes. However, individual genotypes cannot be inferred from low-depth sequencing data without introducing errors. This article presents a new method for inferring an individual's ancestry that takes the uncertainty introduced in next generation sequencing data into account. This is achieved by working directly with genotype likelihoods that contain all relevant information of the unobserved genotypes. Using simulations as well as publicly available sequencing data, we demonstrate that the presented method has great accuracy even for very low-depth data. At the same time, we



# Example of admixture plot from 1000G project

We will use ANGSD (Korneliussen T, 2014) and compute genotype likelihoods on 435 bam-files from the 1000G project.

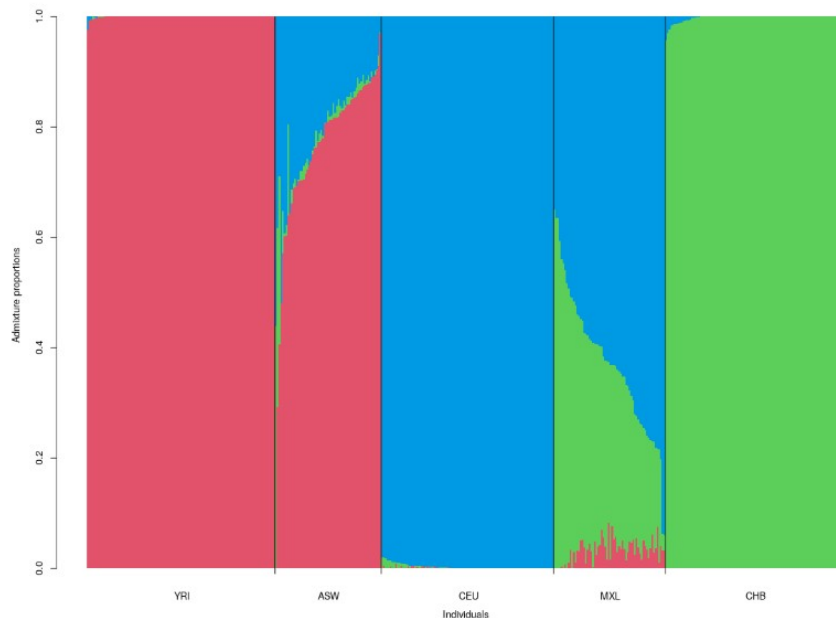
```
1 cd /home/nikolay/Documents/Teaching/PopGen_2023/data/1000G_bam_files
2 find $PWD -name '*.bam' > 1000G_bam_list.txt
3 angsd -bam 1000G_bam_list.txt -GL 2 -doMajorMinor 1 -doMaf 1 -SNP_pval 2e-6 \
4 -minMapQ 30 -minQ 20 -minInd 25 -minMaf 0.05 -doGLF 2 -out 1000G -P 5
```

Next, we will run NGSadmix to infer admixtures proportions:

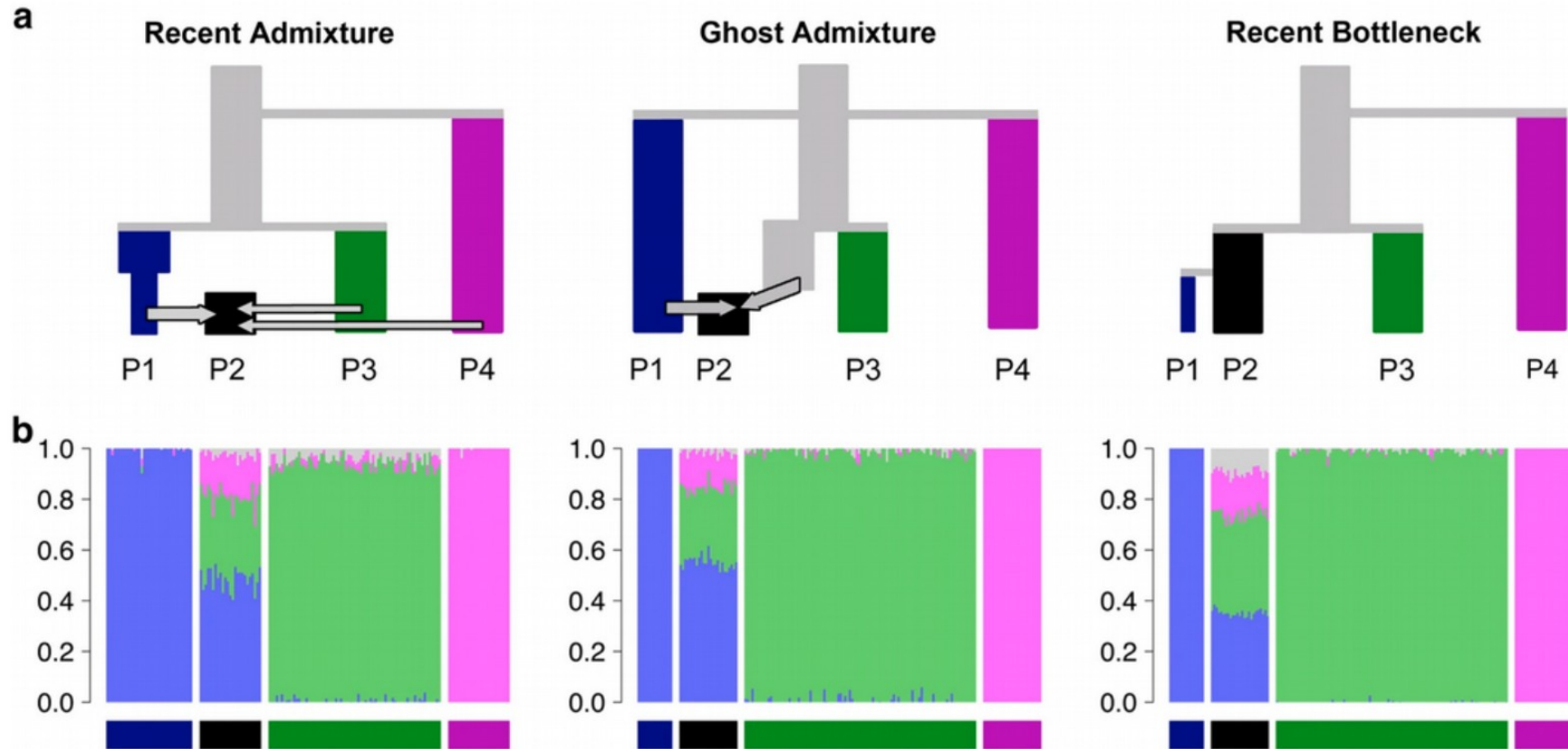
```
1 NGSadmix -likes 1000G.beagle.gz -K 3 -minMaf 0.05 -seed 1 -o 1000G
```

and finally we will plot the results using custom R scripts:

```
1 pops <- readLines("1000G_bam_list.txt")
2 pops <- sapply(strsplit(pops, "\\."), function(x) x[6])
3 source("https://raw.githubusercontent.com/GenisGE/evalAdmix/master/visFuns.R")
4 qopts <- read.table("1000G.qopt")
5 ord <- orderInds(pop = pops, q = qopts, popord = c("YRI", "ASW", "CEU", "MXL",
6 "CHB"))
7 barplot(t(qopts)[, ord], col = c(3, 2, 4), las = 2, space = 0, border = NA)
8 text(sort(tapply(1:length(pops), pops[ord], mean)), -0.05, unique(pops[ord]))
9 abline(v = cumsum(sapply(unique(pops[ord]), function(x) {
10   sum(pops[ord] == x)
11 })), col = 1, lwd = 1.2)
```



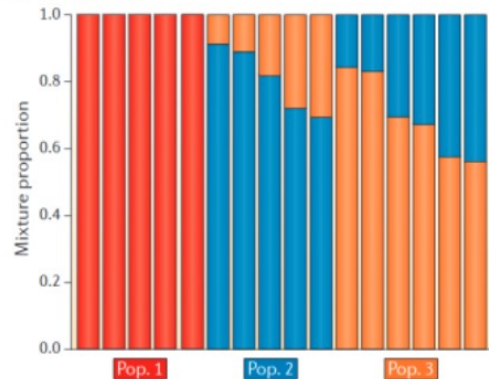
# Admixture bar plots should not be over-interpreted



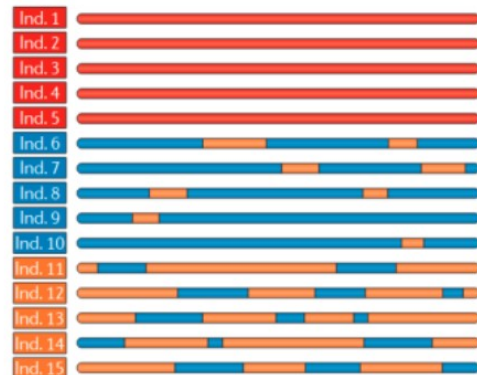
(Lawson DJ, 2018)

# Haplotype clustering

e STRUCTURE



f Chromosome painting



Clustering  
fineSTRUCTURE



Painting Profile

Population structure

- Ancestry assignments in STRUCTURE / ADMIXTURE do not identify where admixture has occurred
- Haplotype-based methods explore **local ancestry**, LD, recombination and subsequently fine-scale patterns of population structure at different scales
- Chromosome painting: the genome is a mosaic of LD blocks separated by recombination
- ChromoPainter and fineSTRUCTURE are tools for resolving **subtle differences** between populations.

(D. Lawson, 2012)

(Schraiber & Akey, 2015)





